

Course	COS 332 S1 2020
Test	Semester test 1 optional rewrite
Started	6/13/20 6:02 PM
Submitted	6/13/20 8:02 PM
Due Date	6/13/20 8:15 PM
Status	Completed
Attempt Score	27 out of 40 points
Time Elapsed	2 hours, 0 minute out of 2 hours
Results Displayed	Submitted Answers, Correct Answers

• **Question 1**

0 out of 0 points

Do you promise not to consult any prohibited resources while writing this test?

Selected Answer:  True

Correct Answer:  True

• **Question 2**

0 out of 1 points

For this and other multiple choice questions in the paper, select the alternative that fits the question best in each case.

The routing prefix in an IPv6 address (normally) consists of the ... of the address.

Selected Answer:  First 32 bits

Correct Answer:  First 64 bits

• **Question 3**

0 out of 1 points

Which of the following considerations is the most important for BGP (Border Gateway Protocol)?

Selected Answer:  Nature of routes inside organisations

Correct Answer:  Routing policies

• **Question 4**

1 out of 1 points

Most layer 3 protocols accomplish routing by

Selected Answer:  Consulting routing tables compiled by layer 7 routing applications.

Correct Answer:  Consulting routing tables compiled by layer 7 routing applications.

• **Question 5**

1 out of 1 points

A protocol is

Selected Answer: 

A set of agreed-upon rules for message interchange.

Correct Answer: 

A set of agreed-upon rules for message interchange.

- **Question 6**

0 out of 1 points

The body in charge of assignment of IP addresses to African ISPs is

Selected Answer: 

ZADNA

Correct Answer: 

AfriNIC

- **Question 7**

1 out of 1 points

Which ISO OSI layer is generally expected to 'throttle' network communication on a busy network to ensure stability of the network?

Selected Answer: 

4

Correct Answer: 

4

- **Question 8**

1 out of 1 points

HTTP responses, Internet messages (RFC2822 / RFC822), and many other protocols use the same mechanism to separate the header and body of the message. They are separated by

Selected Answer: 

A blank line.

Correct Answer: 

A blank line.

- **Question 9**

1 out of 1 points

Which of the following is not a valid UTF-8 sequence?

Selected Answer: 

F2 B3 7F BA

Correct Answer: 

F2 B3 7F BA

- **Question 10**

1 out of 1 points

The routes used by EGPs (external gateway protocols) are typically determined by

Selected Answer: 

Policies

Correct Answer: 

Policies

- **Question 11**

1 out of 1 points

A user is sending mail from jack@up.hill to jill@down.hill (where we assume hill is a valid TLD. Which of the following SMTP commands will be used as part of the SMTP sequence of commands?

Selected Answer: 

rcpt to: jill@down.hill

Correct Answer: 

rcpt to: jill@down.hill

- **Question 12**

1 out of 1 points

In the DNS, SOA refers to

Selected Answer: 

Start of Authority

Correct Answer: 

Start of Authority

- **Question 13**

1 out of 1 points

Which field does not occur in a UDP header?

Selected Answer: 

Sequence number

Correct Answer: 

Sequence number

- **Question 14**

1 out of 1 points

Flow control in TCP uses

Selected Answer: 

Window advertisements

Correct Answer: 

Window advertisements

- **Question 15**

1 out of 1 points

Suppose a TCP segment is transmitted from host A to host B. Suppose host B acknowledges receipt of the segment. However, suppose the message containing the acknowledgement is lost before reaching A. Suppose that no subsequent communication occurs between A and B for a relatively long period. A timeout will then cause

Selected Answer: 

A to retransmit the segment to B.

Correct Answer: 

A to retransmit the segment to B.

- **Question 16**

1 out of 1 points

LDAP is an application layer protocol that

Selected Answer: 

Enables Internet users to find information (such as telephone numbers and email addresses) about individuals.

Correct Answer: 

Enables Internet users to find information (such as telephone numbers and email addresses) about individuals.

- **Question 17**

1 out of 1 points

A zone transfer refers to the process of

Selected Answer: 

Copying DNS entries from a primary server to a secondary server.

Correct Answer: 

Copying DNS entries from a primary server to a secondary server.

- **Question 18**

1 out of 1 points

If the 'do not fragment' flag of an IP datagram is set, and the MTU (maximum transmission unit) of a layer 2 protocol that it needs to traverse is less than the size of the IP datagram

Selected Answer: 

The IP datagram will be discarded.

Correct Answer: 

The IP datagram will be discarded.

- **Question 19**

0 out of 1 points

If a TCP segment is lost, TCP reacts by

Selected Answer: 

Decreasing the rate at which segments are sent.

Correct Answer: 

More than one of the above

- **Question 20**

1 out of 1 points

The file that contains the names, addresses and related information about a domain or subdomain in the DNS is known as a ... file.

Selected Answer: 

Zone

Correct Answer: 

Zone

- **Question 21**

0 out of 1 points

TCP transfers bytes between hosts. Suppose you develop a protocol that works like TCP, but transfers variable sized blocks of data between servers, rather than mere bytes. Each block of data is transferred in its entirety, or the entire block is omitted if any part of the block is not transmitted (nor retransmitted) successfully. After a failure it is entirely possible that subsequent blocks may be transmitted successfully. On which ISO OSI layer does this protocol fit?

Selected Answer: 
3

Correct Answer: 
5

- **Question 22**

1 out of 1 points

Which of the following is not an SNMP request?

Selected Answer: 
Reboot

Correct Answer: 
Reboot

- **Question 23**

1 out of 1 points

The ISO OSI session layer is positioned (exactly) between the ... and ... layers.

Selected Answer: 
Transport, Presentation

Correct Answer: 
Transport, Presentation

- **Question 24**

1 out of 1 points

Which ISO OSI layer's functionality can be informally described as being a 'pipe'?

Selected Answer: 
4

Correct Answer: 
4

- **Question 25**

1 out of 1 points

IPv6

Selected Answer: 
Is already in use on a number of networks.

Correct Answer: 
Is already in use on a number of networks.

- **Question 26**

1 out of 1 points

The three messages sent by TCP when a connection is established, are

Selected Answer: 
SYN, SYN+ACK, ACK

Correct Answer: 

SYN, SYN+ACK, ACK

- **Question 27**

1 out of 1 points

Assume processes A and B are communicating using TCP. For the sake of this question assume that all actions are atomic; when a message is sent, it is received essentially immediately at the other end. Stated differently, two events will never overlap, with one exception - see events 3 and 4 below. Assume A has just sent a message to B with SEQ=100, LEN=500, ACK=1000 and WIN=2000. Immediately after that B sent a message to A with LEN=300 and WIN=1000. Then the following events happen in sequence outlined below. However, events 3 and 4 happen at exactly the same moment and these messages both arrive at their destinations a split second later.

A sends a message to B with LEN=200.

B consumes 500 bytes from its buffer.

A sends a message to B with LEN=600.

B sends a message to A with LEN=200.

A consumes 1000 bytes from its buffer.

A sends a message to B with LEN=300

Below events 1, 3, 4 and 6 will be referred to as messages 1, 3, 4 and 6, respectively.

What is the SEQ value sent along with message 1?

Selected Answer:  600

Correct Answer:  600

Answer range +/- 0 (600 - 600)

- **Question 28**

1 out of 1 points

What is the ACK value sent along with message 6?

Selected Answer:  1500

Correct Answer:  1,500

Answer range +/- 0 (1500 - 1500)

- **Question 29**

1 out of 1 points

What is the ACK value sent along with message 4?

Selected Answer:  800

Correct Answer:  800

Answer range +/- 0 (800 - 800)

- **Question 30**

1 out of 1 points

What is the window advertisement value sent along with message 1?

Selected Answer:  1700

Correct Answer:  1,700

Answer range +/- 0 (1700 - 1700)

- **Question 31**

1 out of 1 points

What is the ACK value sent along with message 3?

Selected Answer:  1300

Correct Answer:  1,300

Answer range +/- 0 (1300 - 1300)

- **Question 32**

0 out of 1 points

Your organisation uses the network address 192.168.201.0/24. (For the purposes of this question assume it is a public IP address whether or not it is actually public.)

However, your organisation has grown to such an extent that it needs a larger address space; in fact you now need the ability to address up to 800 hosts. You speak to the ISP and they indicate that the address you have is part of a larger (supernetted) block that has been allocated to them. Moreover, the blocks from 192.168.177.0/24 to 192.168.220.0/24 are not allocated to customers yet, and they are willing to provide you with a supernet that includes your current block (192.168.201.0/24) that will be able to accommodate the 800 hosts you now need to handle. However, they will provide the smallest such supernet that meets your needs.

Your new address will be of the form 192.168.x.y/z.

Use this scenario to answer this question, as well as the following four questions.

What is the value of z?

Selected Answer:  24

Correct Answer:  22

Answer range +/- 0 (22 - 22)

- **Question 33**

0 out of 1 points

The broadcast address on your (supernetted) network will be of the form 192.168.v.w.

What is the value of v?

Selected Answer:  201

Correct Answer:  203

Answer range +/- 0 (203 - 203)

- **Question 34**

0 out of 1 points

What is the value of x?

Selected Answer:  201

Correct Answer:  200

Answer range +/- 0 (200 - 200)

- **Question 35**

0 out of 1 points

What is the netmask your organisation will use now?

Selected Answer:  255.255.255.0

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
 Exact Match	255.255.252.0	

- **Question 36**

0 out of 1 points

What is the value of y?

Selected Answer:  255

Correct Answer:  0

Answer range +/- 0 (0 - 0)

- **Question 37**

1 out of 1 points

In this question and the four questions that follow, provide your answers as follows

Answer "how many" questions in decimal;

Answer questions that ask for byte sequences as a sequence of 2-digit hexadecimal values, separated by spaces (for example 12 A2 03); and

Answer questions about a Unicode character using a four digit hexadecimal number, without the U+ (for example 01AB, rather than U+01AB).

Case of letters is not important.

Consider the following byte sequence representing chatecters encoded in UTF-8:

C4 94 CE 9E E3 82 A1 31 E2 88 89

How many Unicode characters are present in this string?

Selected Answer:  5

Correct Answer:  5

Answer range +/- 0 (5 - 5)

- **Question 38**

0 out of 1 points

Represent U+0114 as a UTF-8 byte sequence.

Selected Answer:  C494

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	C4 94	

- **Question 39**

0 out of 1 points

Represent U+039E as a UTF-8 byte sequence.

Selected Answer: ✖ CE9E

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	CE 9E	

- **Question 40**

1 out of 1 points

What is the value of the second Unicode character in the UTF-8 string provided above? U+____

Selected Answer: ✔ 039E

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	039E	

- **Question 41**

0 out of 1 points

Represent U+0419 as a UTF-8 byte sequence.

Selected Answer: ✖ D099

Correct Answer:

Evaluation Method	Correct Answer	Case Sensitivity
✔ Exact Match	D0 99	

- **Question 42**

0 out of 0 points

Have you cheated while writing this test?

Selected Answer: ✔
False

Correct Answer: ✔ False

Sunday, April 11, 2021 6:19:17 PM SAST